

Did high speed rail accelerate the development of tourism economy? – Empirical analysis from Northeast China

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ABSTRACT

At present, scholars hold two opposite views on the spatial impact (i.e., positive and negative impact) of high-speed rail (HSR) on tourism economy. To better explore the spatial impact of HSR on tourism economy, this paper uses the difference-in-differences (DID) to empirically test the causal effect of HSR and tourism economy, specifically by using the panel data of cities in Northeast China from 2010 to 2018 as research sample. On the whole, empirical results show that HSR has become an important factor in promoting the development of tourism economy. The impact of HSR on tourism economy in different years indicates that with the continuous promotion of HSR, the effect of HSR-driven tourism economy has become increasingly evident. On this basis, this paper classifies cities according to different types and finds that compared with ordinary prefecture-level cities, central cities are more affected by HSR. Compared with cities with high abundance of tourism resources, cities with low abundance of tourism resources are more affected by HSR. Therefore, current article is convinced that while seizing the opportunity of HSR, cities should actively avoid the negative impact brought by the development of HSR and truly promote the balanced development of regional tourism.

1. Introduction

As is known to all, resources, capital, technology, market, labour force, and the environment are the six major factors affecting industrial development (Shen et al., 2020), among which market openness is the key to industrial development (Fallah et al., 2012). Moreover, the key to market opening generally depends on transportation conditions (Arbués et al., 2015). The transformation of transportation modes is bound to cause a series of spatial impacts on industrial development. Therefore, the opening of HSR and industrial development have become popular issues (Wang et al., 2016).

Up to now, the operation mileage of HSR in China has reached the first in the world, basically realizing the interconnection of all provincial capitals and cities with a population of 500,000, covering 90% of the country's population, China has fully entered the era of HSR (Wang et al., 2016). HSR has the advantages of speed, safety, punctuality and stability, which greatly reduces the travel time cost and expands the commuting space (Urena et al., 2009; Behrens and Pels, 2012). However, different from ordinary railway transportation, HSR mainly

undertakes the task of passenger transportation. Therefore, the role of HSR in regional development is mainly realized by affecting passenger flow, which leads the impact of HSR on tourism industry with large demand for tourists more obvious (Chen and Haynes, 2015).

With the networking of HSR, the development of tourism in different cities or regions shows a heterogeneous performance: positive impact and negative impact. Wherein, the positive effects of HSR on tourism occupy the mainstream. They believed that the operation of HSR substantially reduces the travel cost of tourists (Yan et al., 2014; Gao et al., 2019) and strengthens the development of regional tourism (Delaplace et al., 2014). Moreover, the space-time compression effect produced by HSR has relatively promoted the rapid development of the short-term tourism market, and the development models of one-day and two-day tourism will gradually prevail (Wang et al., 2016, 2019; Felsenstein and Fleischer, 2003). However, other people have also found that although HSR relatively improves travel costs and travel convenience, it relatively intensifies three-dimensional competition among tourism destinations and squeezes the tourist market of cities with rather insufficient tourism resources (Su and Wall, 2009; Wang et al., 2019).

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Undoubtedly, with the development of HSR network in China, urban tourism development may exhibit two different performances, and local and regional factors may be important reasons for this differentiation problem (Yang and Fik, 2014). At present, research on HSR mainly focuses on developed regions, and scholars generally find that HSR has a greater impact on developed regions than on underdeveloped regions. For underdeveloped regions, whether this conclusion is applicable needs further empirical analysis. Based on this, we believe that research on the impact of high-speed rail on tourism development needs to pay more attention to specific economic and geographical regions, such as underdeveloped and remote areas with unique tourism resources.

Northeast China is located in the hinterland of Northeast Asia, it is an important window for China to open up to the north and a central hub for cooperation in Northeast Asia. Its unique advantages provide a foundation for the development of tourism. At the same time, it has abundant tourism resources, and tourism brands with Northeast characteristics such as ice and snow tourism, summer vacation tourism, self-driving tourism, and border tourism are deeply rooted in people's hearts (Jin et al., 2020). Especially in recent years, with the continuous promotion of the strategy of revitalizing Northeast China, vigorously developing tourism has become the top priority to break the bottleneck of economic development in Northeast China (Li et al., 2016). At present, with the gradual improvement of the HSR network, tourism development in Northeast China ushered in unprecedented development opportunities. However, compared with other regions in China, there are still many gaps in tourism development in Northeast China. Therefore, in the context of comprehensively revitalizing the old industrial base in Northeast China, has the opening of HSR truly accelerated the development of tourism economy? Has the impact of HSR on the tourism economy produced different effects in different regions? And how to deal with the impact of HSR construction on tourism development? The exploration and resolution of these issues are of great value for exploring the efficient development path of the tourism economy in Northeast China. At the same time, it will help to enrich the theoretical achievements of regionalization in the research on the impact of HSR on tourism development, and deepen the understanding of the relationship between transport infrastructure and tourism development.

In this paper, we use the opening of HSR in Northeast China as a quasi-natural experiment and empirically examine the impact of HSR on the regional tourism economy by employing the multi-period difference-in-differences (DID) model. The contribution of the current work can be summarized from the following three perspectives. Firstly, we focus the research horizon of HSR's impact on the tourism economy on underdeveloped regions and systematically evaluate the net effect of HSR on tourism economic development in Northeast China, which helps to enrich the existing research perspective. Secondly, we introduce a parallel trend test, placebo test, PSM-DID test, instrumental variable test, and heterogeneity treatment effect test in our econometric assessment framework. These methods can overcome potential endogeneity problems and enhance the robustness, making the research conclusions more scientific and convincing. Thirdly, we analyze the regional heterogeneous effects of HSR on tourism economy by grouping different types of cities. The empirical results have significant practical value and guiding significance for better utilizing the function of HSR and promoting the high-quality development of tourism economy in Northeast China.

The rest of the paper is structured as follows. Section 2 is the literature review and theoretical basis, mainly clarifying the driving system of tourism development and the supporting role of HSR in tourism development. Section 3 is the model construction and data source. Section 4 is the research results. Section 5 is the conclusions and policy implications.

2. Literature review

2.1. The driving mechanism of tourism development

Tourism development is an interactive system composed of tourism consumption and tourism product attraction, and connected by intermediary system and development support system (Peng, 1999; Ma et al., 2021), the dynamic system of tourism development is illustrated in Fig. 1.

People's demand for tourism is the original driving force of tourism development. Tourism consumption mainly involves demand structure and demand level, which are closely related to the population scale and consumption level within the scenic spot market. Tourism products, commonly referred to as resource systems, involve material and non-material tourism attractions. Material tourism attraction mainly refers to all types of landscapes and facilities. It is the basic carrier of tourism activities and material basis for tourism development (Peng, 1999; Majewska, 2015; Campa and López, 2019). The attraction of non-material tourism is related to local cultural characteristics. Urban administrative functions, the leading position of scientific and technological developments are concentrated reflections of a city's image and urban economic activities. It is a direct driving factor in attracting merchants to pursue business and business activities, and also an indirect driving force for the development of tourism economy. Tourism development support system also involves two categories: hard and soft environments. Hard environment mainly refers to regional hardware construction (Li et al., 2010) and involves regional environment, road landscape, urban architecture, street decoration, landscaping, cultural activities, public facilities, and service facilities. Hard environment is a window for the city to express itself, a visual image of the city, and also an important condition for tourism development. By contrast, soft environment means that tourism development needs a soft environment for tourism development with social stability, economic norms, civil civilization, and smooth operation (Cheng et al., 2022). Tourism intermediary system is an intermediate link between products and consumption. This system promotes products to markets, guides consumption, and transforms tourism demand into tourism behaviour. It mainly involves travel agencies or tourism transportation connecting tourist destinations and tourist sources (Vickerman, 2017).

The preceding analysis indicates that transportation, as a link between tourist destinations and tourist sources, is one of the important driving factors of tourism economic development, and its reform is bound to affect the development of tourism economy. As a transformation of transportation mode, HSR has become the preference of travelers for its convenient, fast, and efficient characteristics (Masson and Petiot, 2009). It can be inferred that the rapid development of HSR will have an important spatial impact on tourism economy.

2.2. The influences of HSR on the tourism development

As for the influences of HSR on tourism development, a review of the literature has indicated that research hotspots have mainly focused on the compression effect of HSR on travel space (Wang et al., 2016, 2019; Albatal and Fageda, 2016), change of tourism behaviors (Pagliara et al., 2015), improvement of tourism revenue (Gao et al., 2019; Kang et al., 2021), impact on regional tourism spatial structure (Albatal et al., 2017).

The compression effect of HSR on tourism traffic time is the focus of discussion. For the development of the tourism industry, the most direct impact of HSR is accessibility change (Liu and Zhang, 2018; Wang et al., 2019; Chen and Liu, 2020). Accessibility, as one of the important indicators of transportation network structure and distribution (Wang et al., 2014; Levinson, 2012), is the basis for revealing the compression effect of HSR on the development of the tourism industry. For example, scholars have found that after the opening of TVG, the first HSR in France, the commute time from Paris to Lyon was reduced from 4.5 h to

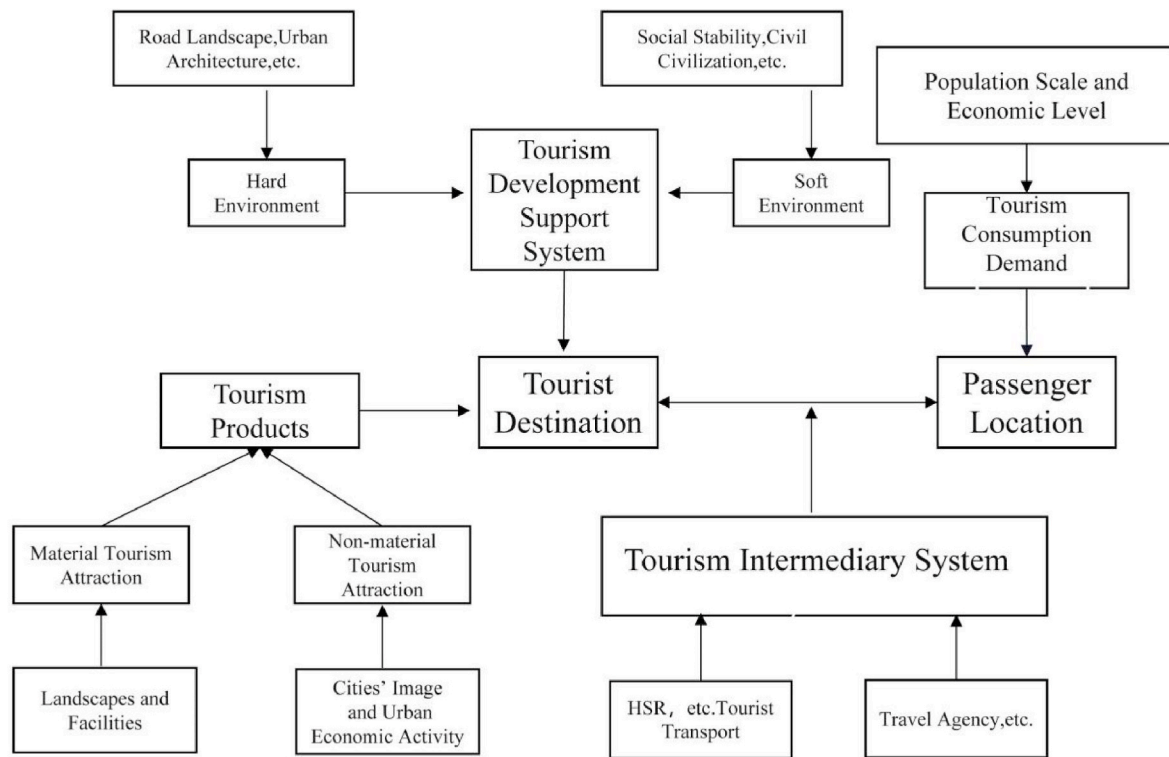


Fig. 1. Dynamic system of tourism development.

about 2 h; after the Shinkansen HSR in Japan was opened, the commute time from Tokyo to Osaka was reduced from the original 7 h–4 h. As for the impact of HSR on tourism behaviors, scholars have found that the space-time compression effect of HSR has significantly changed tourism behaviors and tourism modes. HSR operation makes the “same city effect” evident (Liu and Zhang, 2018). Regional tourism presents an integrated development pattern. Travelers no longer blindly take spatial distance as an evaluation standard when choosing their destinations. After the opening of HSR, the development mode dominated by intra-city tourism and peripheral tourism will gradually change, and inter-provincial and inter-regional tourism mode will gradually prevail (Pagliara et al., 2015). Moreover, the round-trip advantage of HSR also drives the prevalence of one-day tour, weekend tour, short-term tour, and other tourism modes (Wang et al., 2019). High-end business travel, which is an important mode of transportation for medium and high-income groups, also presents high demand for HSR. Regarding the impact of HSR on tourism revenue, some studies have found that the space-time compression effect brought by HSR has greatly stimulated the interaction between tourist destinations and tourist sources, thereby promoting the rapid development of tourism economy. For example, the operation of HSR in California has led to the rapid growth of regional customers, which in turn has promoted the development of regional catering, accommodation, retail and other industries. In the UK, after the coverage of HSR, the central and western regions of the country have gained considerable benefits, and provided more jobs for the region, solving the practical problem of the employment of a large number of laborers in the region. In France, one-fifth of the domestic tourism revenue comes from the French HSR line. Its construction has reduced the distance between Lyon and the capital, and promoted the development of business and leisure tourism in Lyon. Apart from this, HSR will also have an impact on the spatial pattern of tourism market development, and the spatial impact of HSR on the tourism market is different in varying regional scales. HSR trunk roads have a considerable impact on the tourism spatial pattern of large-scale regions (Wang et al., 2012). Meanwhile, intercity express system has a substantial impact on

mesoscale regions, such as urban agglomeration and metropolitan areas, and HSR stations have a considerable impact on tourism space within a city (Zhao et al., 2015; Wang et al., 2020). During the formation period of HSR trunk road, while promoting the formation of transportation tourism economic belt, the spatial polarization characteristics of tourism market development are easily aggravated, thereby relatively limiting the development of global tourism. With the construction and layout of HSR branch line, the network layout of HSR has been preliminarily formed. The latest research results indicated that the network layout of HSR has gradually weakened the siphoning effect and significantly enhanced the diffusion effect, which is conducive to promoting the formation of a global tourism pattern (Wang et al., 2019).

Based on previous research, it is not difficult to find that the impact of HSR on tourism economy presents obvious spatial heterogeneity. Cities along or surrounding the HSR line will gain greater development momentum. But cities far from the HSR faces a certain siphon effect. Therefore, on the whole, after the operation of the HSR, urban tourism economy faces two trends: growth or recession. On the one hand, the opening of HSR has substantially changed the space-time distance between tourist destinations and tourist sources, and the space-time compression effect has considerably promoted the spatial connection among tourism industries (Yan et al., 2014; Gao et al., 2019). Some industries gather along HSR, and the spatial layout presents an evident traffic corridor effect, prompting the formation of a tourism HSR economic belt, which promotes the development of urban tourism economy. On the other hand, the location advantages of different scenic spots have been redefined, scenic spots relatively distant from HSR are subject to relative siphon and crowding out effects, thereby relatively impacting the development of the tourism economy (Felsenstein and Fleischer, 2003).

In summary, a series of discussions have been conducted on the relationship between high-speed rail and tourism economy. Related studies have continuously promoted the construction of the theoretical system for HSR tourism. However, there are still many issues that need to be addressed in current research. On the one hand, research on HSR

tourism mainly focuses on large-scale space and developed regions, while attention to small-scale and underdeveloped regions is generally insufficient. Compared with developed regions, there are significant differences in the organizational forms of regional spatial elements in underdeveloped regions, and it remains to be further revealed whether they are similar to developed regions in terms of HSR tourism effects. On the other hand, most of the current studies rely on the influence of HSR on the tourism economy, and lack of more objective measurement analysis to testify. At the same time, current academic research is mostly based on the positive impact of HSR on tourism development. In response, this research takes Northeast China, an underdeveloped region, as an empirical case, and scientifically evaluates the impact of HSR on tourism economic development based on DID and a battery of robustness tests. Through detailed data and quantitative evidence, this study extends the relevant research on the effect of HSR and regional tourism development, which can provide a broader understanding of the problems of HSR and tourism economy.

3. Methodology

3.1. Econometric model

To evaluate the impact of HSR on tourism economy, it is necessary to compare the results before and after the opening of HSR. In this paper, the opening of HSR is regarded as a quasi-natural experiment, and the difference-in-differences model (DID) is used to test the net impact of HSR on tourism economy.

However, considering that there are certain differences in the opening time of HSR in different cities, we refer to the research of Zhou et al. (2023) and Deng et al. (2019) to further construct a multi-period DID model. The Hausman test showed that the original hypothesis should be rejected, so a fixed-effect model is used. The benchmark model is constructed as Eq. (1).

$$UTI_{it} = \alpha_0 + \alpha_1 HSR_{it} + \gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it} \quad (1)$$

Among them, i and t represent different cities and years respectively, and the explained variable UTI_{it} represents the tourism economy of city i in year t , which is measured by the urban tourism economic income; HSR_{it} is the dummy variable of whether the city has opened the HSR, that is, if the city i has opened HSR in year t , the value is 1; otherwise, the value is 0; X_{it} is a set of control variables used to control the impact of other factors on the urban tourism economy; μ_i , η_t represents the city fixed effect and the year fixed effect, respectively; ε_{it} represents the random disturbance term; α_0 is the constant term, γ is the marginal influence coefficient matrix of the control variable. At the same time, in order to control the influence of autocorrelation, heteroskedasticity, etc. in the model, this paper clusters the standard errors to the city level. The coefficients α_1 in the above model are the focus of our attention, as they measure the net impact of the opening of the HSR on the tourism economy. If the estimated value α_1 is significantly positive, it means that the tourism economy of cities with HSR is significantly improved compared to cities without HSR.

3.2. Variable description

3.2.1. Dependent variable

We select total urban tourism incomes (UTI) as a proxy variable for tourism economic development. Urban tourism revenue, which is a pivotal indicator of tourism economic development and one of the most concerned indicators in existing research (Gao et al., 2019), includes domestic tourism incomes and inbound tourism revenues.

3.2.2. Core explanatory variable

In this paper, we choose HSR opening as the core explanatory variable and set it in the form of a dummy variable. Specifically, in and after the year of the HSR opening of a city, the HSR value is 1, otherwise the

value is 0. It should be noted that, according to the definition of the State Railway Administration of China, high-speed railroads are newly built railroad lines with train speeds of 250 km/h or more, as well as passenger railroads with initial operating speeds of not less than 200 km/h. Therefore, this study follows the above principles to collect and organize information on the opening of high-speed railways in the Northeast region.

3.2.3. Control variables

In order to reduce the estimation errors caused by missing variables, we refer to previous studies, and further control a set of variables related to the development of urban tourism economy. We first controlled the size of the urban population. Population size is an important indicator to measure the degree of urban population aggregation, the intensity and connection of human activities, which can represent the influence of urban population aggregation and market potential on tourism economy (Albalade et al., 2017; Albalade and Fageda, 2016), we used the urban population (UP) of each city to represent it. Second, we controlled for the tourism service capacity variable. Tourism service capacity is the basis to support the development of regional tourism economy, we use the number of employment in accommodation and catering (EAC) to measure. Third, we controlled for variables related to urban public expenditures, which measure the ability of local governments to fund and provide public infrastructure to promote tourism and organize cultural or sports activities that attract tourists (Gao, et al., 2019). According to the existing data, we selected three variables: urban fixed-asset investment (FI), number of urban parks (NP) and urban green area (GA). Finally, we controlled for variables related to urban traffic capacity, including number of trams in operation (NT) and urban paved road area (UR). Because the accessibility of the local traffic is also an important factor affecting the tourist experience and tourism consumption. Table 1 reports the description of the main variables and descriptive statistics for different groups.

Table 1
Description of main variables and descriptive statistics for different groups.

Variables	Description	The treatment group		The control group	
		Mean	Std. Dev.	Mean	Std. Dev.
Dependent variable					
UTI	Urban tourism economic income	5.049	4.464	1.700	1.805
Core explanatory variable					
HSR	= 1 if a city opened HSR in year t and afterwards; = 0 otherwise	0.667	0.474	0.000	0.000
Control variables					
NP	The logarithm of the number of urban parks	3.107	0.919	2.451	0.573
GA	The logarithm of the urban green area	8.730	0.871	7.934	0.819
NT	The logarithm of the number of trams in operation	7.219	1.061	6.177	0.633
EAC	The logarithm of the number of employees in accommodation and catering	8.115	1.538	7.185	0.770
FI	The logarithm of the fixed-asset investment	16.26	1.021	15.46	0.865
UP	The logarithm of the urban population	5.986	0.530	5.335	0.420
UR	The logarithm of the urban paved road area	7.484	0.897	6.572	0.694
Observations	Total = 306	108	108	198	198

3.3. Data source

In order to measure the change of tourism economy in Northeast China before and after the opening of high-speed railway, this paper conducts an empirical analysis based on the panel data of 34 prefecture-level cities in Northeast China from 2010 to 2018. Our sample selection was mainly based on the completeness and availability of the data. By the end of 2018, a total of 12 cities in the study area had opened high-speed railways, forming the treatment group of this paper, while the remaining samples formed the control group. Among them, the socioeconomic data are mainly from the China Urban Statistical Yearbook and the city statistical bulletin of each prefecture-level city, the tourism data are from the China Tourism Statistical Yearbook and the official website of each municipal government, and the high-speed rail opening data are from the official website of the China Railway Customer Service Center (www.12306.cn) and Baidu Encyclopedia.

It should be noted that the empirical research carried out in Northeast China is mainly based on the following considerations: Compared with other regions in China, the construction of HSR in Northeast China started earlier, and the rapid construction of HSR in Northeast China from 2012 to 2018, HSR line in Northeast China has been relatively stable, and there is not much volatility. HSR in Northeast China has experienced two periods: formation of trunk roads and improvement of branch roads. The Harbin-Dalian HSR was officially started and officially opened in December 2012. As a major passenger transport artery running through the core economic area of Northeast China, the Harbin-Dalian trunk line is significant in improving traffic accessibility of Northeast China and optimizing passenger transport structure and regional tourism spatial network in this region. Thereafter, it entered the improvement period of the branch line, and the Panjin-Yingkou HSR commenced operation in September 2013. The Harbin-Qiqihar, Shenyang-Dandong, and Dandong-Dalian HSRs also started operation in August, September, and December 2015, respectively. With the gradual improvement of the branch network, the networked layout of HSR in Northeast China is highlighted. Considering the time lag effect of HSR on tourism economy, this paper is carried out in areas where the has been relatively stable. However, the construction of HSR in other regions of China is still in a period of rapid fluctuation, which has a certain impact on empirical research.

4. Results

4.1. Parallel trend test

A basic assumption of using DID method to identify policy effects is that the tourism economy characteristics of the treatment group and the control group meet a common trend, i.e., before the opening of HSR, the trend of tourism economy over time in the treatment and control groups is parallel, and there is no significant difference. Therefore, we depict the average change trend of urban tourism economic income in the treatment group and the control group from 2010 to 2018. As the results in Fig. 2, before HSR was opened, the change trends of the treatment group and the control group were basically the same. Since the first HSR opened in Northeast China in 2012, the trend of the two curves has been significantly differentiated, and the gap between the treatment group and the control group has gradually widened, which shows that the treatment group and the control group selected in this paper meet the parallel trend hypothesis of DID. At the same time, in order to make this conclusion more convincing, we further replace the time dummy variables used in the benchmark model with the dummy variables of each year in the study period, to examine the parallel trend from a more rigorous perspective. It can be seen from Fig. 3, the estimated coefficients of each period before the opening of HSR were not significant, which means that there is no significant difference in the changing trend between the control group and the treatment group before the opening of HSR, which again indicates that the sample passed the parallel trend

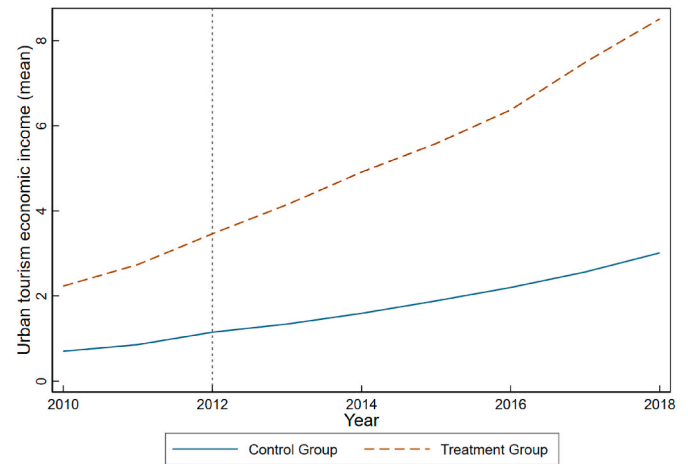


Fig. 2. Evolution trend of urban tourism economic income in treatment group and control group.

test.

4.2. Benchmark regression

4.2.1. Static economic effects

This section mainly uses the DID method to evaluate the impact of HSR on tourism economy. Results are reported in Table 2. To avoid the influence of control variables on the results, we implement both DID regressions without adding control variables and introducing all control variables, and the results are shown in columns (1) and (2) of Table 2, respectively. All estimation results indicate that the spatial effect of HSR on tourism economic development in Northeast China is consistent and significantly positive at the 5% level, regardless of whether control variables are included in the model, implying that the opening of HSR in Northeast China effectively promotes the development of tourism economy. This finding verifies that the HSR tourism effect also exists in underdeveloped regions. From the regression results of control variables, the estimated coefficients of the number of urban parks (NP) and urban green areas (GA) are significantly positive, indicating that the improvement of urban public expenditure capacity may promote the development of tourism economy. This conclusion is also supported by the study of Gao et al. (2019). The regression coefficient of employment in accommodation and catering (EAC) was significantly positive at the level of 10%, indicating a positive correlation between urban tourism service capacity and the development of tourism economy. The coefficient of urban paved road area (UR) is also significantly positive, which means that urban public transportation can promote the development of tourism economy by improving the accessibility and satisfaction of tourists, which is in line with the research conclusions of Prideaux (2000) and Delclòs-Alió et al. (2022). The number of trams in operation (NT), the fixed-asset investment (FI) and urban population (UP) have no significant impact on the tourism economic development.

4.2.2. Dynamic economic effects

Table 2 shows the results of the average impact of HSR on tourism economy within a specific period, and the estimated results are static. Considering the stage of the development of HSR network, the impact of HSR on tourism economy presents stage differences. In order to further test the dynamic impact of HSR on the development of tourism economy at different time points, we draw on the research of Ma et al. (2021) and Cai and Ye. (2020) to construct the following dynamic effect models. The regression equation is as follows:

$$UTI_{it} = \alpha_0 + \sum_{j=-3}^6 \alpha_j HSR_{it}^j + \gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it} \quad (2)$$

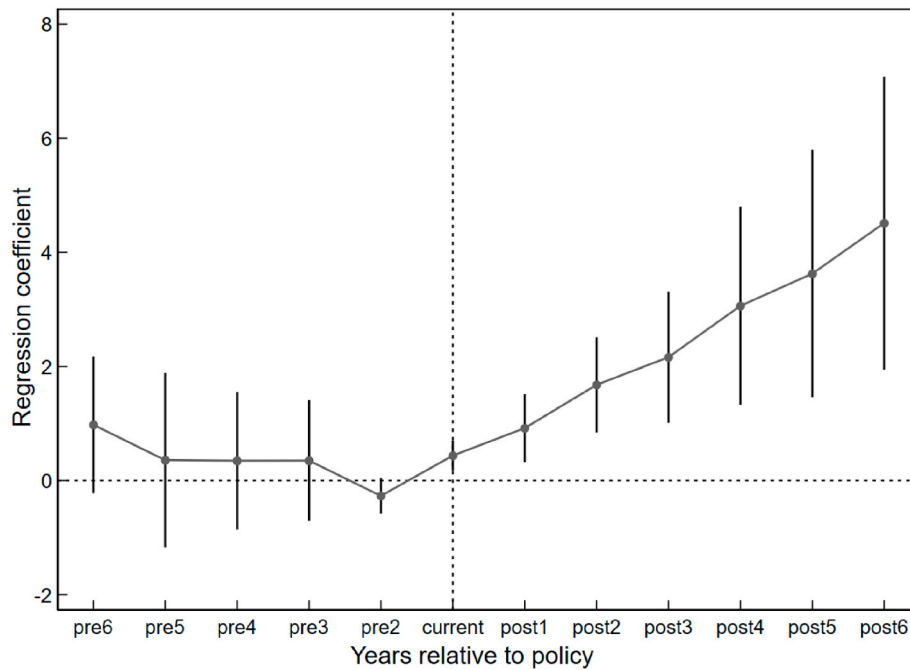


Fig. 3. Parallel trend test. **Notes:** The regression coefficients of each period are the relative value with the base period, the circle dots are the estimation of the coefficient point of each period, and the short vertical lines are the lower 95% confidence interval of the standard error calculation at the city level.

Table 2

Baseline regression results.

Variables	UTI	
	(1)	(2)
HSR	1.838** (2.61)	1.790** (2.56)
NP	–	0.742* (0.66)
GA	–	0.806** (0.86)
NT	–	–0.076 (–0.12)
EAC	–	0.271* (1.35)
FI	–	–0.169 (–0.36)
UP	–	10.577 (1.43)
UR	–	0.144* (0.42)
Constant	1.241*** (5.28)	–66.004 (–1.58)
City FE	YES	YES
Year FE	YES	YES
Observations	306	306
R ²	0.542	0.594

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Among them, HSR_{it}^j is the dummy variable in the j year ($j = -3, -2, \dots, 6$), $HSR_{it}^j = 1$ means that t year is the j th year after the opening of HSR in city i , otherwise it is 0. The regression coefficient α_j represents the impact of HSR on the tourism economy in the j th year before and after the opening of HSR.

As shown in the regression results in Table 3, the estimated coefficients are statistically significant after the opening of HSR, regardless of whether control variables are added or not, indicating that the opening of HSR significantly contributes to the growth of the tourism economy. Note that after the opening of HSR, the coefficient shows a steady upward trend. The possible reasons are as follows. On the one hand, HSR effect has evident time lag characteristics, thereby making the growth of tourism economy driven by HSR present lag characteristics. On the other hand, with the gradual improvement of the HSR network, the superposition effect of HSR on tourism economy growth appears, making the regression coefficient increase significantly.

Table 3

Dynamic test results.

Variables	UTI	
	(1) Add the control variables	(2) Control variables not added
HSR_3	0.500 (0.66)	0.532 (0.71)
HSR_2	0.322 (0.51)	0.341 (0.50)
HSR_1	–0.267 (–1.52)	–0.241 (–1.44)
HSR0	0.426** (2.73)	0.399** (2.72)
HSR1	0.942** (2.60)	0.952*** (3.18)
HSR2	1.699*** (3.40)	1.697*** (3.59)
HSR3	2.177*** (3.18)	2.169*** (3.36)
HSR4	3.072*** (3.00)	2.970*** (3.39)
HSR5	3.640*** (2.84)	3.866*** (3.20)
HSR6	4.523*** (2.98)	4.670*** (3.16)
NP	0.275** (0.32)	–
GA	0.884 (1.25)	–
NT	–0.077 (–0.11)	–
EAC	0.228* (1.14)	–
FI	0.211 (0.42)	–
UP	7.487 (1.30)	–
UR	–0.146 (–0.37)	–
Constant	–51.895 (–1.65)	1.258*** (5.53)
City FE	YES	YES
Year FE	YES	YES
Observations	306	306
R ²	0.698	0.670

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. HSR_3, HSR_2 and HSR_1 are the third, second and first years of before the opening of the HSR; HSR0 is the year of the opening of HSR; HSR1–HSR6 is the 1–6 years after the opening of HSR.

4.3. Impact of HSR on tourism economic development of different types of cities

From the benchmark regression results, it can be seen that HSR can promote the development of tourism economy, but due to differences in city levels and tourism resources, the impact of HSR on tourism economy in different cities may be different. In order to explore this problem, first of all, we divide the research samples into central cities and ordinary cities according to different city levels, the central cities are Dalian,

Shenyang, Harbin and Changchun, and the rest are ordinary cities. Specifically, since all of the central cities have already opened HSR, when examining the effect of HSR in the central cities, we delete the sample of ordinary cities that have opened HSR and take the cities that have not opened HSR as the control group. Secondly, we classify cities into rich-resource and deficient-resource cities according to their tourism resource quality scores. The cities with tourism resource quality scores above the median are rich-resource cities, and the rest are deficient-resource cities. The results of heterogeneity regression are shown in Table 4.

4.3.1. Urban scale effect

Columns (1) and (2) of Table 4 demonstrate the results of city scale heterogeneity. The regression results show that after adding the control variables, the impact of HSR on central cities is more significant, whereas that on ordinary cities is not significant, which means that the opening of HSR has a stronger pulling effect on the tourism economy of the center city. Central cities are more affected by HSR, which may be caused by the following reasons. On the one hand, central cities have certain economic strength and development potential, and the demand for tourism economic development is high. Small and medium-sized cities have limited economic strength and lack a driving force for the development of urban tourism economy. Second, central cities are more attractive to tourism consumers. The construction of HSR has accelerated the inflow of tourism consumers. The development of such cities is relatively mature. Accordingly, the opening of HSR easily gives full play to their tourism advantages and drives the development of tourism economy.

4.3.2. Urban tourism resource effect

Columns (3) and (4) of Table 4 demonstrate the results of city tourism resource heterogeneity. The regression results show that after adding the control variables, the impact of HSR on deficient-resource cities is more significant. This result may be related to the fact that

Table 4
Heterogeneity test results.

Variables	UTI			
	Urban scale		Urban tourism resource	
	(1) central cities	(2) ordinary cities	(3) rich- resources cities	(4) deficient- resource cities
HSR	4.761*** (9.26)	0.555 (1.25)	1.387 (1.31)	0.975*** (2.97)
NP	0.562 (0.57)	−0.677 (−1.52)	1.103 (0.64)	−0.227 (−1.20)
GA	1.235 (1.27)	1.058** (2.45)	0.873 (0.32)	1.084*** (4.83)
NT	0.054 (0.08)	0.345 (0.71)	−0.179 (−0.13)	−0.042 (−0.15)
EAC	0.370* (1.89)	0.130 (1.08)	1.276 (1.73)	−0.014 (−0.18)
FI	0.080 (0.16)	−0.354 (−0.83)	0.309 (0.49)	−0.709*** (−5.52)
UP	7.429 (1.43)	4.759* (1.98)	52.129** (2.70)	3.334** (2.48)
UR	0.179 (0.51)	0.052 (0.20)	−0.903 (−0.64)	−0.144 (−0.62)
Constant	−56.901* (−1.86)	−29.825** (−2.10)	−328.471** (−2.59)	−12.738** (−2.36)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	234	270	126	180
R ²	0.670	0.660	0.750	0.834

Notes: Robust *t*-statistics in parentheses. ****p* < 0.01, ***p* < 0.05, **p* < 0.1. Urban tourism resources score = $\sum$ the number of tourism resources $\times$ the quality of tourism resources, the values of the quality of tourism resources is 1–5 respectively according to A–5A.

cities with rich tourism resources attract more tourism consumers with their resource advantages, but the traffic drive is not evident. But for the deficient-resource cities, the improvement of location conditions brought by HSR relatively reduces the limitation of insufficient tourism resources and presents a more evident driving effect.

4.4. Placebo test

Although parallel trend tests control for a range of observable factors that change with time going, the benchmark regression results are still disturbed by other unobservable factors that change with time going. Therefore, this study addresses this problem with a placebo test.

First, we advance the opening time of HSR 2 and 3 years earlier, respectively, and construct a false dummy variable (HSR^{false}) as a placebo, to re-examine the impact of HSR on the tourism economy. The test results are shown in columns (1) and (2) of Table 5, and the estimated coefficient of HSR^{false} is not significant, which shows that there is no systematic difference between the time trend of the treatment group and control group, and once again proves that the opening of HSR promotes the development of tourism economy.

Second, this paper conducts placebo experiments by randomly generating treatment group. Specifically, in this paper, 12 cities were randomly selected as false treatment group, and the other cities were randomly selected as false control group. Then the estimated coefficient of the impact of HSR on the urban tourism economy can be obtained through the benchmark model regression after the implementation of placebo. The above procedure is repeated 1000 times, resulting in 1000 spurious estimated coefficients. Fig. 4 plots the probability density distribution of the false estimation coefficient. It can be found that the estimated coefficient after the placebo is evenly distributed around zero and followed a normal distribution. The true estimated coefficient (i.e., the HSR coefficient in column 2 of Table 2) is a clear outlier in the placebo test and meets the expectations of the placebo test. The above results indicate that other unobservable variables have no significant effect on the core estimation results of this paper.

Table 5
Robustness Test results.

Variables	Time-placebo test		Extreme-value samples were removed	
	Two year in advance	Three years in advance	Winsorize1%	Trim1%
	(1)	(2)	(3)	(4)
HSR^{false}	−0.511 (−0.71)	−0.468 (−0.75)	–	–
HSR	–	–	1.666** (2.55)	1.168** (2.53)
NP	0.537 (0.44)	0.557 (0.46)	0.327 (0.39)	0.183 (0.47)
GA	0.667 (0.65)	0.654 (0.63)	0.852 (1.04)	0.732 (1.69)
NT	0.055 (0.08)	0.042 (0.06)	−0.072 (−0.11)	−0.375 (−0.72)
EAC	0.205 (1.01)	0.213 (1.05)	0.229 (1.29)	−0.001 (−0.01)
FI	−0.203 (−0.43)	−0.215 (−0.46)	−0.259 (−0.60)	−0.622** (−2.44)
UP	12.502 (1.49)	12.485 (1.48)	13.545 (1.63)	28.241*** (3.02)
UR	−0.020 (−0.05)	−0.023 (−0.06)	0.110 (0.33)	0.130 (0.40)
Constant	−73.720 (−1.57)	−73.367 (−1.57)	−80.038* (−1.74)	−150.965*** (−2.95)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	306	306	306	272
R ²	0.546	0.546	0.627	0.727

Notes: Robust *t*-statistics in parentheses. ****p* < 0.01, ***p* < 0.05, **p* < 0.1.

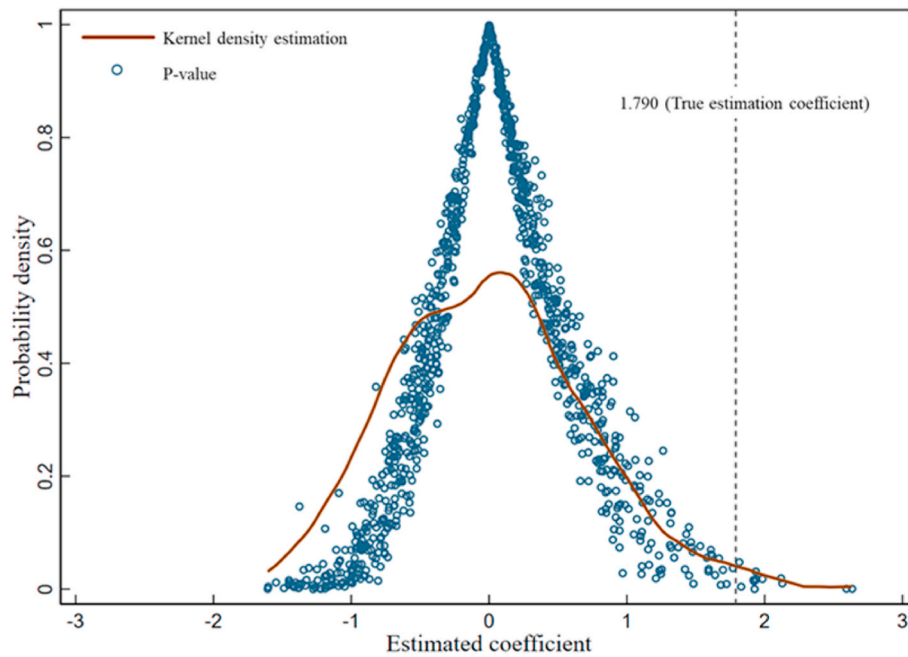


Fig. 4. The results of placebo test. **Notes:** The dashed line indicates the true estimate in the baseline regression.

At the same time, in order to exclude outliers on the benchmark regression results, we winsorize 1% and trim 1% for all continuous variables and reestimated them. The new estimates are reported in columns (3) and (4) of Table 5. Results show that the estimated coefficient of the benchmark model is still significantly positive after excluding the outliers' interference, which further supports the core research conclusions of this paper.

4.5. Results from PSM-DID method

When analyzing the impact of HSR on tourism economy with the use of DID model, the premise needs to meet the assumption of a parallel trend. To some extent, the endogenous interference of time and the fixed effect of urban individuals can be excluded, but this method cannot eliminate the interference of selective bias. The main reason is that the difference between cities is bigger, the government is not completely random when choosing cities to open HSR, and cities with a relatively good level of tourism development have a greater probability of opening HSR. If the endogenous problems caused by such reverse causality are not considered, the evaluation results may be biased. In order to solve the above problems and enhance the reliability of the research results, this paper refers to Li et al. (2021), and applies the PSM-DID model to clarify the causal relationship between HSR and tourism economy.

Specifically, first of all, we use the control variables as the covariates, take whether or not open the HSR as the independent variable, and used the Logit model to regression process the treatment group and the control group, calculate the corresponding score of each sample, and then get the prediction value of the matching score. Secondly, the radius matching method is used to match the treatment group and control group, and construct a control group with the most similar city characteristics to the processing group; Finally, the DID model is used to further identify the net effect of HSR on tourism economy after the propensity score matching.

Fig. 5 and Table 6 show the results of the balance test for PSM matching, which shows that the standardized deviation is less than 10% for all variables after PSM matching. By comparing the results before matching, the standardized deviation reduction of all variables was above 90%, and the results of *t*-test of all variables did not reject the null hypothesis of no systematic difference between the treatment group and

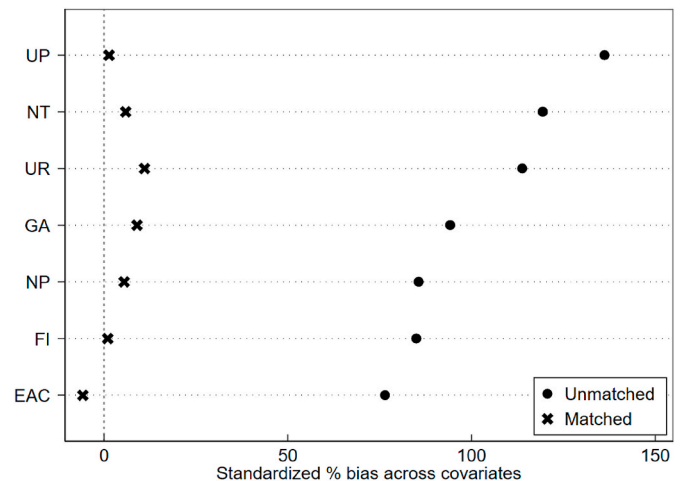


Fig. 5. Standardized % bias of covariates.

the control group. The results show that the matching variables and methods selected in this paper are reasonable and effective.

The regression results of the PSM-DID method are shown in Table 7. We find that the estimation coefficient of HSR is always positive and significant, whether the control variables are added or not, which is consistent with the benchmark regression results, indicating that the opening of HSR significantly promoted the development of tourism economy, which once again confirmed the robustness of the core conclusion of this paper.

4.6. Results from the IV 2SLS method

In order to further avoid the endogeneity problem caused by reverse causality and omitted variable bias, this study used the instrumental variable two-stage least squares (IV 2SLS) approach for further testing. Specifically, first, we select the relief degree of the land surface (*Rdls*) as the instrumental variable of the opening of HSR, mainly because the relief degree of the land surface is directly related to the construction cost of transportation infrastructure (Campos and Rus, 2009), thus

Table 6

The results of the balance test for PSM.

Variable	Unmatched matched	Mean		%reduct		t-test	
		Treated (108)	Control (198)	%bias	bias	t	p > t
NP	U	3.107	2.452	85.6	93.7	7.67	0.000
	M	2.443	2.402	5.4		0.54	0.593
GA	U	8.730	7.934	94.2	90.5	7.95	0.000
	M	8.092	8.016	8.9		0.69	0.492
NT	U	7.219	6.177	119.4	95.1	10.76	0.000
	M	6.435	6.384	5.8		0.53	0.596
EAC	U	8.115	7.185	76.5	92.3	7.05	0.000
	M	7.234	7.305	−5.9		−0.51	0.613
FI	U	16.263	15.459	84.9	98.9	7.28	0.000
	M	15.675	15.666	0.9		0.06	0.950
UP	U	5.986	5.335	136.2	99.1	11.79	0.000
	M	5.563	5.557	1.3		0.13	0.898
UR	U	7.484	6.572	113.8	90.4	9.89	0.000
	M	6.866	6.778	11.0		0.96	0.338

Notes: U stands for unmatched, M stands for matched.**Table 7**

Results of the PSM-DID regression.

Variables	UTI	
	(1) Add the control variables	(2) Control variables not added
HSR	0.811** (2.08)	1.016** (2.22)
NP	−0.635 (−1.46)	–
GA	1.152** (2.75)	–
NT	0.295 (0.61)	–
EAC	0.127 (0.96)	–
FI	−0.159 (−0.38)	–
UP	4.246* (1.99)	–
UR	−0.082 (−0.32)	–
Constant	−29.521** (−2.30)	0.797*** (4.71)
City FE	YES	YES
Year FE	YES	YES
Observations	256	256
R ²	0.673	0.615

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

affecting the direction of HSR lines, satisfying the basic criteria for the relevance of IV 2SLS. In addition, the relief degree of the land surface variable also well satisfy the exogenous requirements. Second, referring to [Martincus et al. \(2017\)](#) and [Baum et al. \(2007\)](#), whether the cities opened railway stations (*Rail*) in 1993 is used as the instrumental variable for the opening of HSR, which is because the idea of building ordinary railways in history is similar to that of building HSR today. Railway cities also need to consider geographical, military and political location factors, so the location of HSR stations is highly related to historical railway lines. At the same time, the historical railway construction is far away from the current time, which will not affect the current development of the urban tourism economy, so it also satisfies the requirements of exogenous and relevance of IV 2SLS. Where the relief degree of the land surface is calculated with reference to the research by [Feng et al. \(2007\)](#), and whether a railway station was opened in 1993 is based on statistics from China's Railway Road Map 1993. It should be noted that since the *Rdls* and *Rail* variables do not change over time, this paper multiplies them with the year dummy variable to obtain the tool variables *Rdls_IV* and *Rail_IV* which change with time going.

The estimation results of IV 2SLS are shown in [Table 8](#). From columns (1) and (3) of [Table 8](#), the estimation coefficient of *Rdls_IV* and *Rail_IV* are significant, which means that the lower the cities with the relief degree of land surface and the cities that historically opened the railway station, the higher the probability of the opening of HSR, thus verifying that the instrumental variables constructed in this paper satisfy the relevance requirement. At the same time, the first-stage F values of IVA and IVB are greater than 10, indicating that there is no weak

Table 8

Estimation results for the IV 2SLS method.

Variables	IVA		IVB	
	HSR	UTI	HSR	UTI
	(1)	(2)	(3)	(4)
HSR	–	1.104*** (3.20)	–	1.956*** (4.01)
<i>Rdls_IV</i>	3.726*** (31.10)	–	–	–
<i>Rail_IV</i>	–	–	0.799*** (13.29)	–
First-stage F value	105.54	–	27.74	–
Control Variables	YES	YES	YES	YES
Constant	−0.400** (−2.05)	−16.165*** (−7.64)	0.287 (0.87)	−15.433*** (−7.29)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	306	306	306	306
R ²	0.854	0.724	0.606	0.729

Notes: Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. IVA and IVB are IV 2SLS regression results for the instrumental variables *Rdls_IV* and *Rail_IV*, respectively. In columns (1) and (3), the dependent variable is HSR, with columns (2) and (4) for *UTI*.

instrumental variable problem. The results in columns (2) and (4) show that the estimation coefficient of HSR is still significantly positive, indicating that the core conclusions of this paper remain robust after excluding potential endogenous problems.

4.7. Heterogeneous treatment effects analysis

[Callaway and Sant'Anna \(2021\)](#) showed that the estimation of the staggered DID model is the weighted average of the effect of each group in each period, depending on the implicit homogeneity processing effect assumption. Due to the heterogeneity of the response intensity of different cities for the opening of HSR in different years in reality, the estimation bias may occur when using the traditional staggered DID model to identify the impact effect of HSR on tourism economy. To ensure the robustness of the two-way fixed effect model, this study refers to the study of [De Chaisemartin and D'Haultfoeuille \(2022\)](#), heterogeneous treatment effects analysis was introduced into a two-way fixed effect model, and the local processing effect of the opening of HSR was tested using `twowayfweights` command in Stata17 and fuzzy DID model. The results of the estimator weights show that of all the 72 wt, 72 wt are positive and 0 are negative, and the standard deviation of the

metric estimate with heterogeneous treatment effects is 2.685, which is much larger than 0, indicating that the estimation results under heterogeneous treatment effects are relatively robust. Furthermore, we use the fuzzy DID model to estimate the local average treatment effect (LATE) for the handling individuals with changes in status (switchers). The indicator variable for handling changes in status are G_T and its first-order forward G_{T1} , the time variable is *year*, and the treatment variable is whether HSR is opened. The test results are shown in Table 9, and the fuzzy DID estimate value for the *Wald-TC* is 0.440, and it is significantly positive at the 1% level. It shows that the LATE of switchers remains significantly present after considering the heterogeneous treatment effects, which again verifies the robustness of the conclusions.

5. Conclusions and policy implications

5.1. Conclusions

The large-scale construction of HSR in China has directly promoted the rapid flow of tourists between tourist sources and tourists. However, does this promotion of the tourism economy by HSR still hold in the Northeast? The answer is of practical importance to profoundly understand the relationship between HSR and tourism economy, to explore the construction options of HSR in special economic geographic regions (e.g., underdeveloped areas), and to explore efficient development paths for tourism. In response, this paper takes the opening of HSR as a quasi-natural experiment and empirically tests the effect of HSR opening on the tourism economy by adopting the DID method based on panel data of 34 prefecture-level cities in Northeast China from 2010 to 2018. The main findings are as follows.

First, we confirm the positive effect of the opening of HSR on the tourism economy of Northeast China. This conclusion is still robust after a series of robustness tests such as parallel trend test, placebo test, PSM-DID test, instrumental variable test, and heterogeneity treatment effect test. Meanwhile, the dynamic effect test results show that the impact of HSR on the tourism economy is continuously significant as the HSR network continues advancing, and this promotion effect is becoming increasingly obvious.

Second, similar to the developed regions, we find that the impact of HSR on the tourism economy of cities in Northeast China shows significant regional size heterogeneity. Compared with small and medium-sized cities, the central cities have a good foundation for tourism economic development and have stronger dynamics and potential for rapid tourism development. The impact of HSR on the development of tourism in central cities is more obvious. This further proves the robustness of the conclusion that the impact of high-speed rail on developed cities is greater than that of less developed cities.

Third, the richness of urban tourism resources is also an important factor affecting the tourism effect generated by HSR in Northeast China. The more scarce the city tourism resources, the more obvious the promotion effect of HSR on the city tourism economy.

On average, our findings for Northeast China are largely similar to those of previous literature examining the national scope, confirming that the positive effects of HSR opening on the tourism economy are widespread. At the same time, our study also provides some new and interesting findings. For example, our study indicates that the positive effects of HSR on the tourism economy tend to continue to strengthen as the HSR network in Northeast China continues to improve, which affirms the long-term benefits that the construction of HSR in the Northeast possesses. We also find that HSR has a more significant impact on the tourism economy of large-scale cities such as central cities in

Northeast China. Clearly, this heterogeneous impact provides a warning of the regional imbalance in the construction of HSR in the Northeast. Our study also shows that HSR generates better tourism effects in deficient-resource cities in Northeast China, which finding provides a valid insight and reference for future HSR network planning priorities and tourism economy development directions in Northeast China. Although our analysis focuses on Northeast China, the results and lessons learned can be extended to other similarly underdeveloped regions in China and around the world, which share similar regional characteristics and face similar tourism development issues.

5.2. Policy implications

Our findings have important policy implications for HSR planning and urban tourism development. As for HSR planning. First of all, the layout of HSR lines and the setting of HSR stations should not only consider the geological conditions, natural environment, population scale and other factors, but also fully consider the spatial distribution characteristics of industries with a large demand for population flow, so as to improve the maximum use efficiency of HSR. Second, we should continue to speed up the construction of HSR branch lines and improve the layout of rapid transportation lines. At the same time, the future development trend of the HSR network should be combined to obligate a certain strategic development space for the development of HSR in the edge and peripheral areas. To meet the needs of the edge cities for HSR development.

As for the development of urban tourism. First of all, the opening of HSR relatively promotes the development of urban tourism economy. Therefore, tourism routes and high-quality projects should be built along the HSR lines in the future, and inter-city tourism cooperation and resource sharing must be strengthened. Second, although the development of HSR has strengthened the rapid development of tourism economy in central cities, it may relatively squeeze the development of tourism economy in ordinary cities. Therefore, a metropolitan rapid transportation tourism circle should be built around the four central cities with the combination of point, line, and area, to further radiate and drive the development of surrounding cities. Third, this paper also proves that the impact of HSR will be more significant for cities with insufficient tourism resources. Therefore, there is a need to speed up the construction of other HSR lines or branch lines, and bring more tourist cities and scenic spots into the HSR commuter circle.

We summarize the limitations of this study. First, due to the limited availability of data, we only considered the impact of HSR opening on tourism economic income in Northeast China, while exploring the deeper impact of HSR on urban tourism development quality and development efficiency may be better for solving the current problems. Therefore, the results can be extended in the future with more comprehensive data from the perspectives of tourism development quality and tourism efficiency. Second, this study doesn't deeply explore the channels through which the impact of HSR on tourism economy is realized. Future research can reveal the mechanism of the impact of HSR on urban tourism economy from the perspective of mediating effects. Finally, due to space limitations, this study only considers the impact of HSR on the tourism economy in Northeast China. However, many other cities in underdeveloped and remote regions of China still exist, and similar studies on these cities in the future can facilitate the enrichment and validation of our findings.

CRedit author statement

Shaobo Wang (First Author): Conceptualization, Writing - original draft, Funding acquisition, Formal analysis, Supervision. Yutao Zhou (Corresponding author): Conceptualization, Investigation, Formal analysis, Writing - original draft, Writing - review & editing, Methodology. Jianke Guo: Visualization, Writing - review & editing, Supervision. Jinhuan Mao: Data Curation, Investigation.

Table 9

Local average treatment effects under heterogeneous treatment effects.

Estimated value	LATE	Std_Err	t	P_value
<i>W_TC</i>	0.440	0.143	3.089	0.002

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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